

CoRoT-2b

R-0214

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240731 · created 2026-07-17T00:33:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460522.7897 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.012
Transit depth (Rp/Rs)^2	3.47 +/- 0.46 [%]
Orbital Inclination (inc)	85.07 +/- 1.39
Airmass coefficient 1 (a1)	1.357 +/- 0.02
Airmass coefficient 2 (a2)	-0.243 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	None
Optimal Aperture	4.07
Optimal Annulus	14.58
Transit Duration (day)	0.0849 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.012 vs 0.1667 (deviation 1.61σ)
Duration fit vs published	0.0849 d vs 0.095 d (deviation 1.23σ)
Mid-time O–C	8.7 min
Depth SNR	15.5
Residual scatter	1.46 %

Light curve

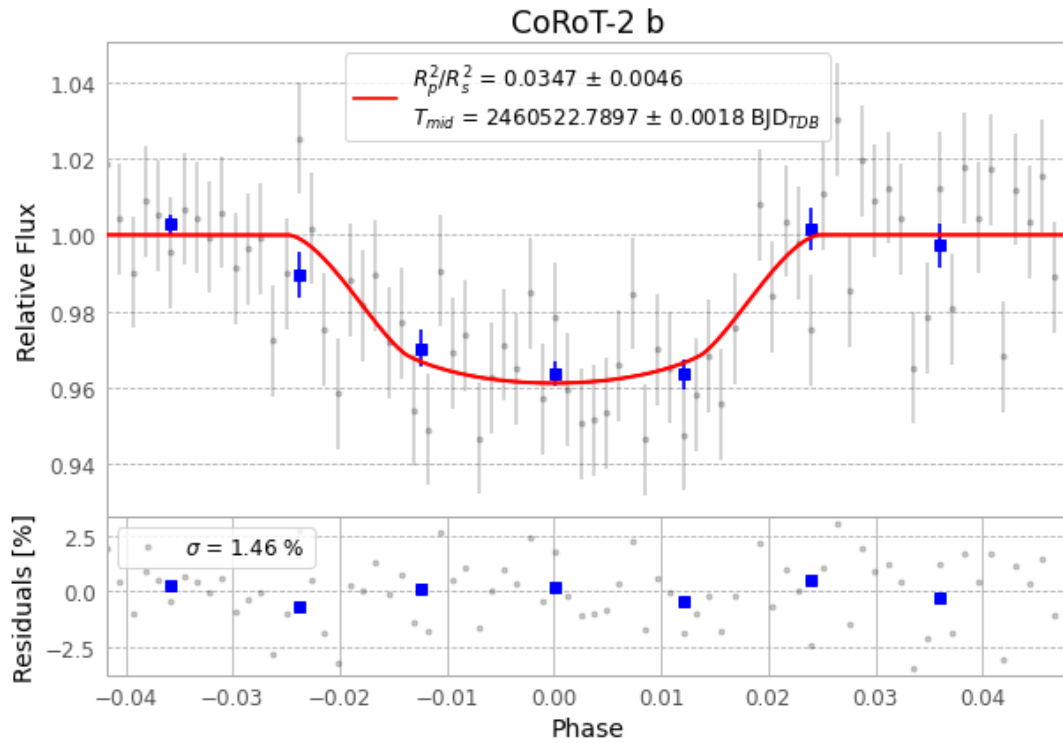


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-07-30.png (SHA-256 f45b5ae0187fcaab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 17b438e982f10a97...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

76 FITS frames (50 MB), CoRoT-2240731050315.FITS → CoRoT-2240731084817.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240731.log (59663368b7a0...), FinalLightCurve_CoRoT-2 b_2024-07-30.png (f45b5ae0187f...), FinalLightCurve_CoRoT-2 b_2024-07-30.csv (4fdf6de25e78...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:33Z.